No.	Status	Date
LNDE-CAL-UT-001 Rev 3	Issued	2023-08-14

Calibration Procedure - UT Thickness Gauges and PAUT Equipment

Scope and Application

This procedure establishes the requirements for the calibration and verification of ultrasonic thickness gauges and Phased Array Ultrasonic Testing (PAUT) equipment at the Hamilton Plant. It ensures that all digital thickness measurements and encoded scans comply with the technical requirements of ASME BPVC Section V. The revision specifically addresses updated calibration protocols for PAUT equipment to maintain accuracy during asset integrity assessments.

The methodology applies to all carbon steel and stainless steel piping systems. For A312 TP316 lines, the equipment must be calibrated specifically at the stainless-steel velocity setting to account for material-specific acoustic properties. All calibration activities must be performed using certified reference blocks with thicknesses traceable to national standards, ensuring that the equipment is capable of detecting wall loss relative to the line t-min.

Responsibilities

David Park is responsible for the technical oversight and approval of this procedure. As the Quality Manager, David Park ensures that all NDE personnel are qualified to SNT-TC-1A UT Level III standards and that the calibration intervals for PAUT equipment are strictly maintained. This oversight includes the verification of calibration records and the validation of automated data acquisition systems used during inspections.

NDE technicians are responsible for performing daily field calibrations and documenting the results in the equipment logs. Any instrument that fails to meet the linearity requirements specified in ASME BPVC Section V must be removed from service immediately. The technician must verify that the equipment is functioning within the specified tolerances before any thickness readings are recorded for the purpose of calculating remaining life.

Calibration Requirements for PAUT and UT

Calibration must be performed at the start of each work shift and whenever there is a change in probe, cable, or operator. For PAUT equipment, this includes a wedge delay calibration and a Sensitivity Equalization (ACG) to ensure uniform signal response across all focal laws. The calibration must be verified against a step wedge that brackets the expected thickness range of the Hamilton Plant assets.

If a reading is found to be below the retirement thickness (t-min), the technician must immediately raise an Non-Conformance Report (NCR). The calibration must be re-verified following any reading that indicates significant wall loss to ensure the data is not the result of instrument drift. All A312 TP316 lines require a dedicated velocity calibration to ensure the accuracy of the ultrasonic path length in austenitic materials.

Data Integrity and Reporting

All calibration data must be recorded and stored alongside the inspection results to facilitate the calculation of corrosion rates. The remaining life is determined by the formula (t - t-min) / rate, where the rate is the larger of the long-term or short-term values. Accurate calibration is essential as the next thickness reading is due at no more than half the remaining life, subject to the class maxima of 5 or 10 years.

The integrity of the calibration ensures that the RBI intervals, such as the 3 years for Class 1 sour gas and regenerator overhead or 6 years for other lines, are based on valid data. David Park will periodically audit the calibration logs to ensure compliance with this procedure and ASME BPVC Section V. Any discrepancies in calibration will require a re-inspection of all components measured since the last successful verification.

Sign-off	Name	Function	Date	Signature
Prepared	David Park	Level III / Quality Manager, SNT-TC-1A UT Level III	2023-08-14	_____